



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: IX
SUBJECT: COMPUTER APPLICATIONS

TOTAL MARKS: 40
DATE: 20.6.22

Instructions:

The Question Paper comprises of Two Parts. Part-I is based on Multiple Choice Questions and Application questions. Part -II is based on Java Programs. The intended marks for questions are given in brackets []. This paper consists of three printed pages.

PART-I (20 marks)
[Attempt all question]

Question 1

Choose the correct answer from the choices given:

[10×1=10]

i) If we want to access a variable from all the classes within a package, we declare a variable as:

- | | |
|------------------------|--------------|
| a) private | b) public |
| c) no access specifier | d) protected |

ii) What is meant by state of an object?

- | | |
|---------------------------|-------------------------------|
| a) Function of the object | b) Data Members of the object |
| b) Content of an object | d) Class of an object |

iii) .CLASS file is a

- | | |
|----------------|-----------------|
| a) Interpreter | b) Compiler |
| b) Byte code | d) Machine code |

iv) The number of relational operators in Java is

- | | |
|------|------|
| c) 6 | b) 5 |
| d) 8 | d) 4 |

v) If a=6, b=5, c=12 then what will be the value of x in the following statement:

$x = (a > b ? (a > c ? 20 : 10) : 15)$

- | | |
|-------|-------|
| a) 10 | b) 20 |
| c) 15 | d) 12 |

vi) Arrange the data types in as ascending order according to the size

PART-II (20 marks)

[Attempt all questions. Each program should be written using Variable descriptions/Mnemonic Codes so that the logic of the program is clearly depicted]

Question 3

[10]

A salesman working in a company “Mehra and sons” gets the commission on the sales done by him depending on the following conditions:

Sales (₹)	Commission (percentage of sales)
Upto ₹10000/-	10 %
₹10001- ₹20000/-	15 %
₹20001 - ₹50000/-	30 %
> ₹50000/-	40 %

Write a program in Java to input the sales of a salesman. Calculate and display the Commission of the salesman.

Question 4

[10]

The Simple Interest (SI) and Compound Interest (CI) of a sum (P) for a given time (T) and rate (R) can be calculated as:

Simple interest is calculated on the principal, rate, time.

1) $SI = p \cdot r \cdot t / 100$

Compound interest is calculated on the principal amount and the accumulated interest of previous periods, and thus can be regarded as “interest on interest.”

2) $CI = P \cdot ((1 + R/100)^T - 1)$

Write a menu driven program in Java using switch case to calculate Simple and Compound Interest. Input sum, rate, time and type of Interest (1 for Simple interest, 2 for Compound Interest). Calculate and display the amount and interest earned.

